

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

Code of Conduct:

I Vignesh S / 192325002 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Vignesh S

DATA INTERPRETATION

Q1) Rising Reward Trend (10 to 45)

RL Concept: Learning curve, policy improvement and convergence of cumulative reward.

Observation: Across episodes the reward climbs steadily from about 10 to about 45 and then flattens.

Interpretation: The rising reward shows the agent is improving its policy through experience: as it updates action values from the rewards it receives, it increasingly selects higher-value actions, so the return per episode grows. The flattening near 45 indicates the policy is converging toward a near-optimal solution where further gains are small.

Justification: In value-based and policy-based methods the expected return increases as estimates approach their true values; improvements are large early and diminish as the policy nears optimality, producing exactly this concave, saturating curve.

Conclusion: The upward-then-plateau trend confirms successful learning and clear policy improvement, with the agent converging to a stable, near-optimal policy.

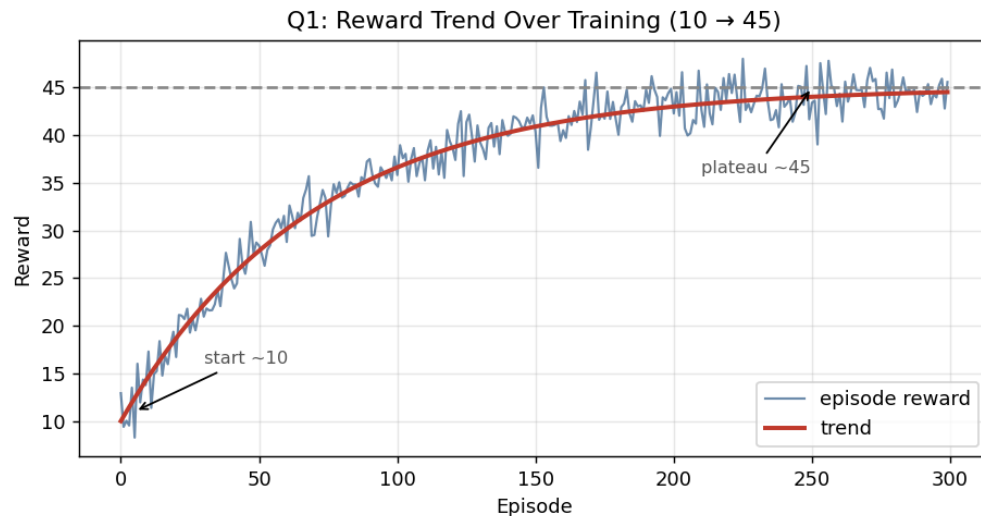


Figure 1: Rising Reward Trend (10 to 45).

Q2) Effect of Epsilon on Cumulative Reward

RL Concept: Exploration versus exploitation controlled by the epsilon parameter of an epsilon-greedy policy.

Observation: Cumulative reward is highest at $\epsilon = 0.1$ and lowest at $\epsilon = 0.5$.

Interpretation: A larger epsilon makes the agent act randomly more often, so even after learning good actions it keeps choosing sub-optimal ones, which lowers cumulative reward. With $\epsilon = 0.1$ the agent mostly exploits its learned knowledge while still exploring just enough, giving the best return.

Justification: Once the high-value actions are known, additional exploration mainly wastes steps on low-value actions; a small epsilon therefore balances the trade-off better than a large one.

Conclusion: $\epsilon = 0.1$ provides the best performance here; excessive exploration at $\epsilon = 0.5$ degrades cumulative reward.

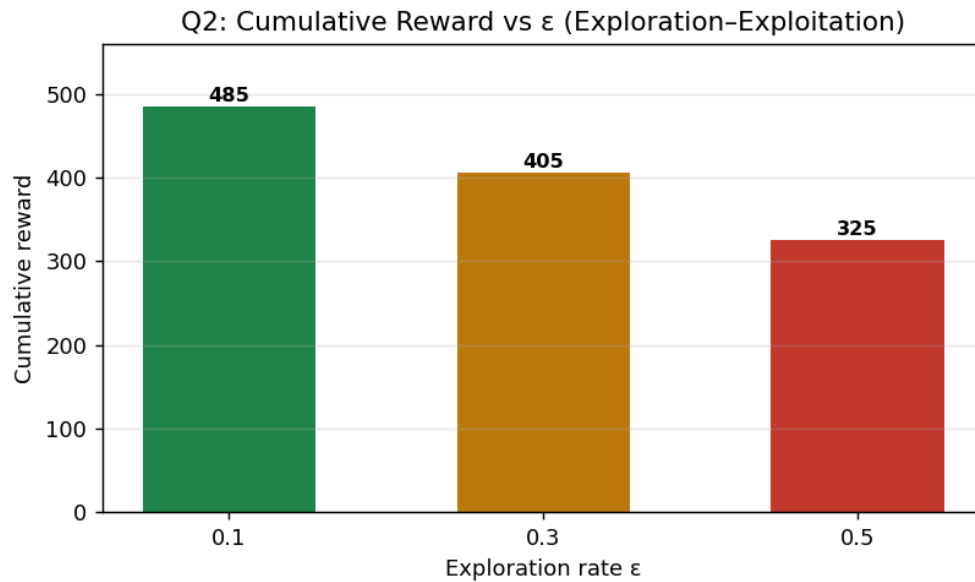


Figure 2: Effect of Epsilon on Cumulative Reward.

Q3) Q-Learning versus SARSA

RL Concept: Off-policy learning (Q-Learning) versus on-policy learning (SARSA).

Observation: Q-Learning gives consistently slightly higher average reward than SARSA at 100, 200 and 300 episodes.

Interpretation: Q-Learning updates toward the greedy optimal action (the maximum next-state value) irrespective of the exploratory action actually taken, so it targets the optimal policy directly and reaches higher reward. SARSA updates toward the action its behaviour policy actually takes, so it learns a safer, more conservative policy that includes the cost of exploration.

Justification: Because Q-Learning bootstraps on the maximum action-value, it propagates optimal-value information faster, which is why its average reward rises above SARSA and it converges faster in this comparison.

Conclusion: Q-Learning converges faster to a higher-reward policy; SARSA is more conservative and slightly lower here, though often safer in risky environments.

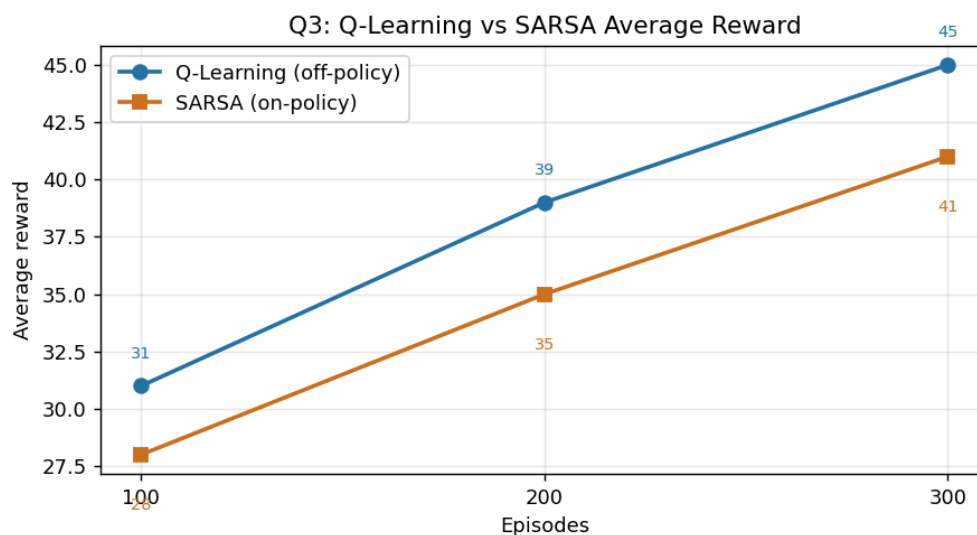


Figure 3: Q-Learning versus SARSA.

Q4) Effect of the Discount Factor on Reward

RL Concept: The discount factor gamma and its role in long-term reward optimisation.

Observation: Average reward increases as gamma rises from 0.2 to 0.5 to 0.9.

Interpretation: A small gamma makes the agent short-sighted, valuing only immediate rewards and ignoring the long-term consequences of its actions. As gamma increases, future rewards carry more weight, so the agent optimises the full long-term return, which raises the total reward it collects.

Justification: The return is a gamma-weighted sum of future rewards; higher gamma lengthens the effective planning horizon, which is essential when good outcomes depend on delayed rewards.

Conclusion: Gamma = 0.9 is the most suitable here for long-term optimisation, though values extremely close to 1 can slow or destabilise convergence.

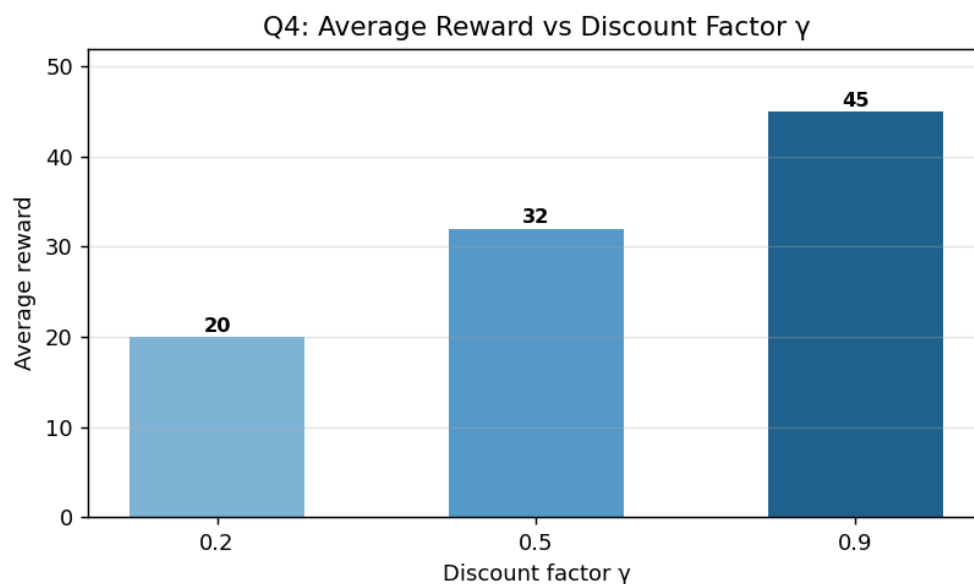


Figure 4: Effect of the Discount Factor on Reward.

Q5) Success Rate Improvement (45% to 92%)

RL Concept: Policy improvement and convergence measured by task success rate.

Observation: Success rate rises from 45 percent at 50 episodes to 92 percent at 200 episodes.

Interpretation: The steep, sustained rise shows the agent is learning to reach the goal far more reliably as it accumulates experience. Reaching 92 percent indicates a near-optimal policy, while the remaining failures suggest either residual exploration or environment stochasticity rather than a fully optimal, deterministic solution.

Justification: A monotonically improving success rate that levels off at a high value is the signature of convergence toward an effective policy; a truly optimal policy in a deterministic task would approach 100 percent.

Conclusion: The trend shows the agent has learned a near-optimal policy; learning is effective, though the 8 percent gap means it is not yet perfectly optimal.

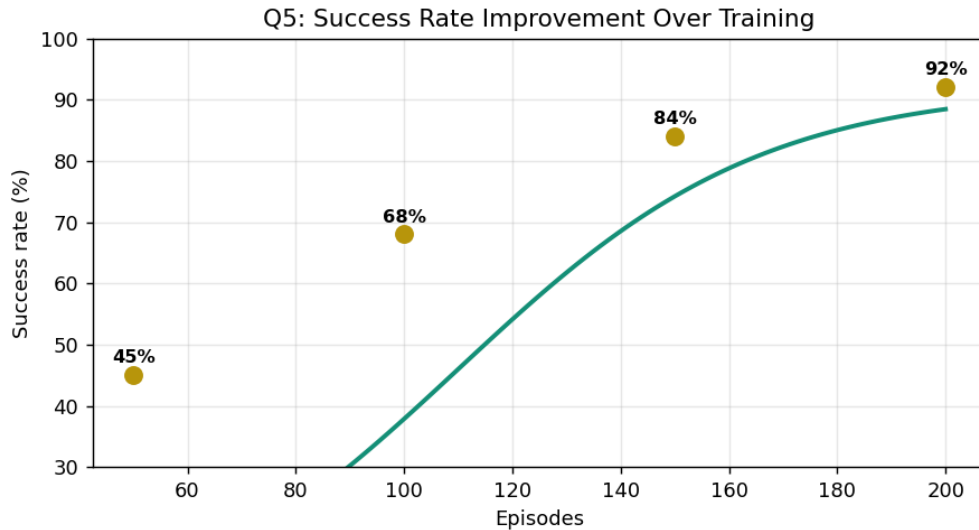


Figure 5: Success Rate Improvement (45% to 92%).

Q6) Learning Rate and Convergence Stability

RL Concept: The learning rate (step size) and its effect on convergence speed and stability.

Observation: The agent is most stable at a learning rate of 0.05; 0.1 is faster but oscillates, and 0.01 is stable but slow.

Interpretation: A high learning rate of 0.1 takes large update steps that overshoot good estimates and cause the reward to oscillate. A low rate of 0.01 takes tiny steps and converges slowly. A rate of 0.05 moves quickly enough to learn fast while remaining small enough to stay stable.

Justification: The learning rate scales each update to the value or policy parameters; too large amplifies noise and diverges, too small wastes samples, so an intermediate value gives the best speed-stability trade-off.

Conclusion: A learning rate of 0.05 offers the best trade-off between convergence speed and stability in this CartPole experiment.

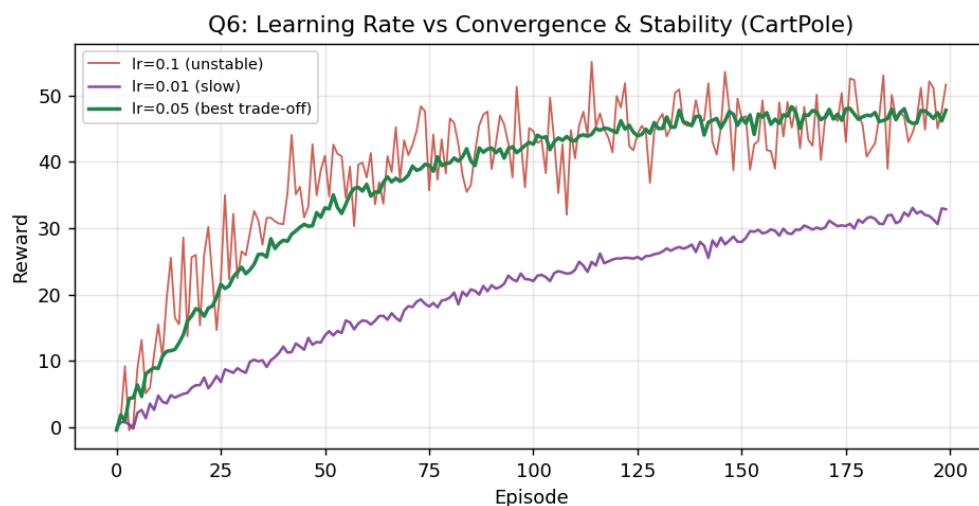


Figure 6: Learning Rate and Convergence Stability.

Q7) Comparison of Exploration Strategies

RL Concept: Exploration strategies: epsilon-greedy, softmax and random exploration.

Observation: Cumulative reward is highest with epsilon-greedy, moderate with softmax and lowest with random exploration.

Interpretation: Random exploration never exploits what has been learned, so it keeps taking poor actions and earns the least. Softmax explores in proportion to estimated value and does better. Epsilon-greedy exploits the best-known action almost all the time with only occasional exploration, which maximises cumulative reward when a clearly best action exists.

Justification: In a setting with a well-separated optimal action, strong exploitation is rewarded; epsilon-greedy concentrates on the best action while retaining minimal exploration, whereas random selection ignores learned values entirely.

Conclusion: Epsilon-greedy achieves the best performance in this scenario because it balances exploitation of the known best action with just enough exploration.

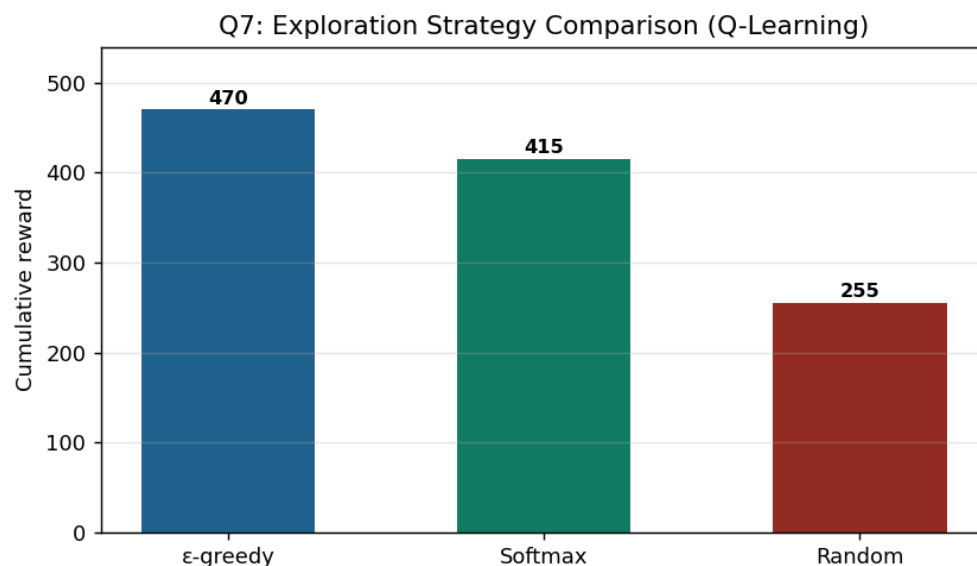


Figure 7: Comparison of Exploration Strategies.

Q8) Steps-to-Goal Reduction in a Maze

RL Concept: Convergence to an optimal (shortest-path) policy.

Observation: The number of steps to reach the goal falls from 50 initially to 15 after 300 episodes and then stays flat.

Interpretation: Fewer steps means the agent has found a more efficient route to the goal. The steady decrease shows it is progressively pruning unnecessary moves, and the plateau at about 15 steps indicates it has reached the shortest feasible path the environment allows.

Justification: As reward and value information propagate back from the goal, the greedy policy increasingly follows the shortest path; when step count stops decreasing, the policy has converged.

Conclusion: The reduction in steps and the plateau together indicate convergence to a near-optimal, shortest-path policy.

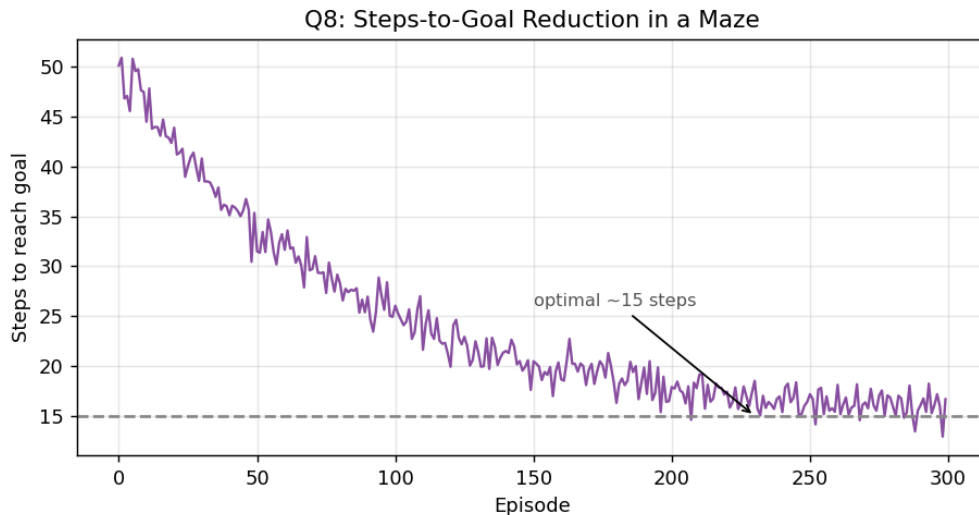


Figure 8: Steps-to-Goal Reduction in a Maze.

Q9) SARSA with Different Discount Factors

RL Concept: The discount factor in SARSA and the balance between immediate and future rewards.

Observation: The SARSA agent achieves the highest long-term reward with $\gamma = 0.95$, above 0.7 and 0.4.

Interpretation: With $\gamma = 0.4$ the agent heavily discounts the future and behaves myopically, so it misses rewards that come later. Raising γ to 0.95 makes distant rewards significant, so the agent plans over a long horizon and accumulates more long-term reward.

Justification: A discount factor near 1 lengthens the effective horizon, which is exactly what a task dominated by delayed rewards requires; lower values sacrifice long-term gains for short-term ones.

Conclusion: $\gamma = 0.95$ is most effective for long-term reward because it weights future outcomes strongly while still keeping the return finite and learnable.

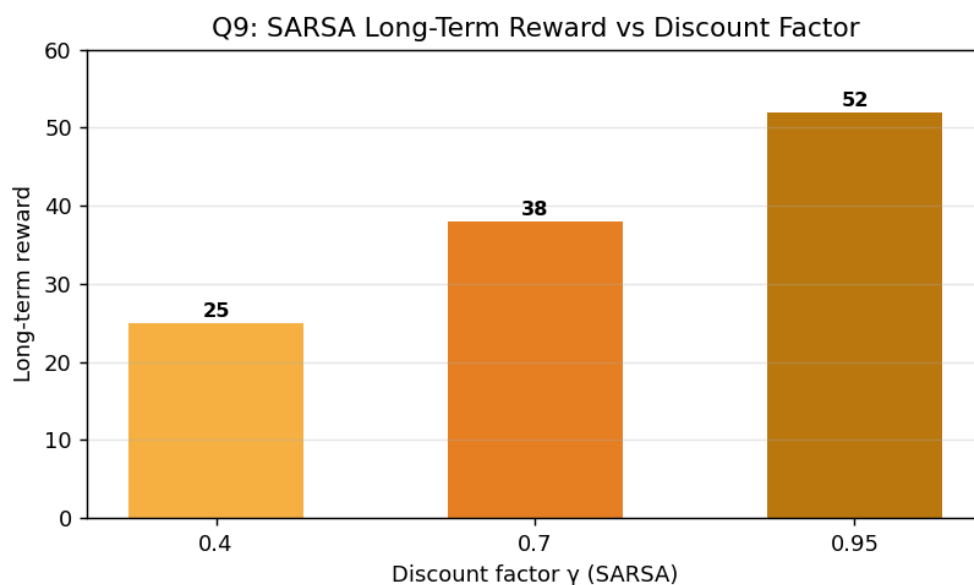


Figure 9: SARSA with Different Discount Factors.

Q10) Reward Stabilisation in a Stochastic Environment

RL Concept: Convergence and robustness of learning under environmental stochasticity.

Observation: Rewards fluctuate widely in the early episodes but settle to a stable level after about 500 episodes.

Interpretation: Early volatility comes from both exploration and the random transitions of the stochastic environment, so returns vary from episode to episode. As the agent gathers enough samples to average out this randomness and its exploration decays, the value estimates converge and the reward stabilises, showing a robust policy.

Justification: Estimating expected returns in a stochastic setting needs many samples to reduce variance; a curve that stabilises after sufficient episodes indicates the policy has converged and is robust to the environment's randomness.

Conclusion: The move from fluctuating to stable reward demonstrates genuine policy improvement and robust learning, confirming the agent has converged despite the stochastic environment.

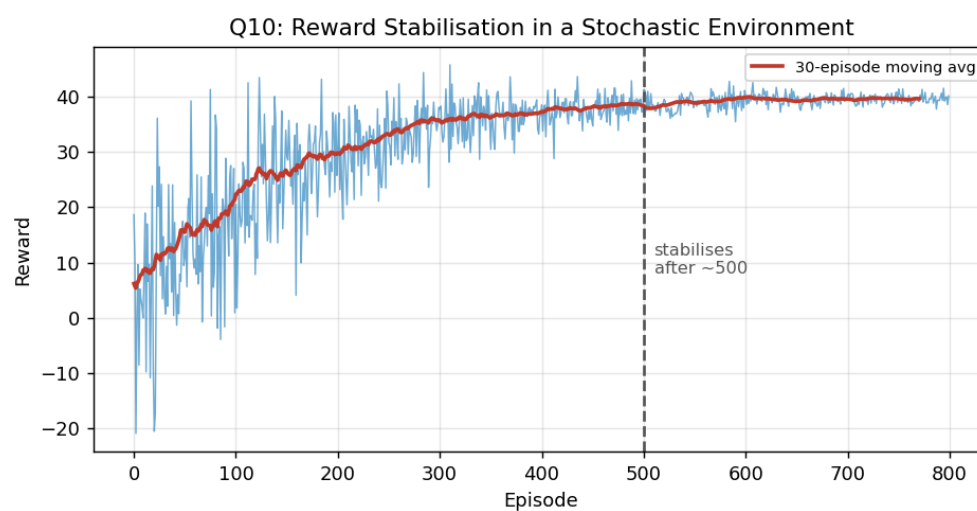


Figure 10: Reward Stabilisation in a Stochastic Environment.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data

Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand